

Precision and Recall

Project: Responsible Plant Classifier

1. Your mission

Your learning target

I can choose the denominator that answers the question.

I can calculate precision and recall with a declared positive class.

I can report undefined ratios honestly.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

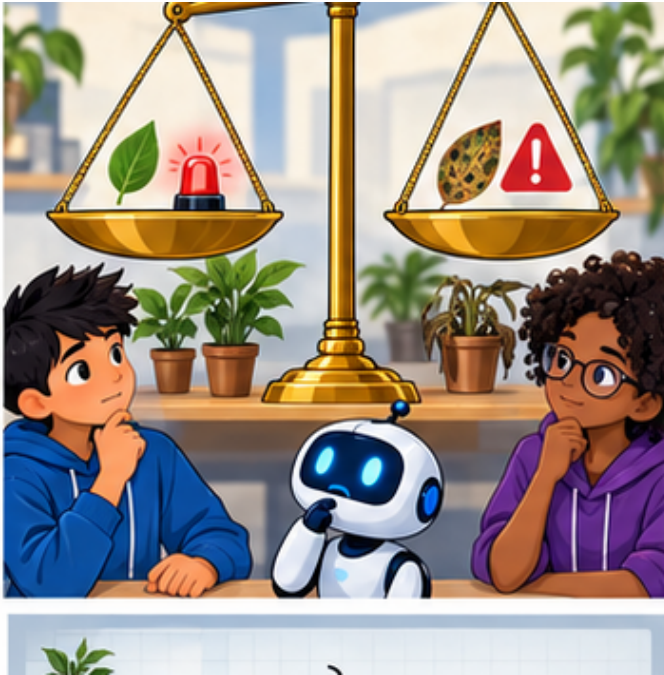
Without opening the old lesson, explain one key idea from Accuracy and Imbalance. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Scores and Calibration Intuition. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Precision measures how many positive predictions are right; recall measures how many actual positives are found.



LEARNING TARGET TP=9,FP=3,FN=6,TN=12. Compute precision and recall. What happens to precision when a system predicts no positives?

Builder name and experiment date:

Readiness for the connected explanation

Why can 80 percent accuracy hide missed blocked cases?

This week's end goal

By the end, I can explain precision and recall using a traced example and a stated limitation.

2. Notice the evidence

PICTURE ENTRY

Notice → **Predict** → **Explain**

Begin with visible evidence; do not explain more than the picture and information support.



MYSTERY The team must use precision and recall to solve a model-building problem. What information do they have? What important evidence is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what evidence would confirm or reject your prediction.

My prediction and supporting evidence:

A closer explanation

Blocked remains positive. Precision asks: of the cases flagged blocked, how many were actually blocked according to the reference? Recall asks: of all reference-blocked cases, how many were found? Both use true positives, but they answer different questions. Keep reference rows and prediction columns ordered clear then blocked.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building process; this week examines precision recall and denominators.

3. See how it works

ML MECHANISM

Input → Model → Evidence

IN

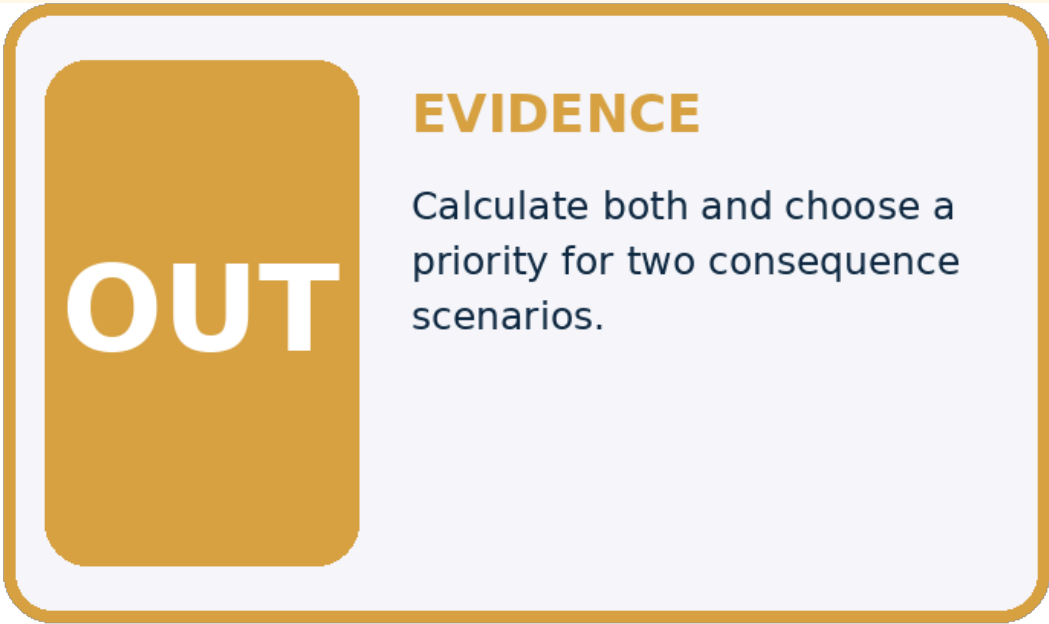
INPUT

Calculate both and choose a priority for two consequence scenarios.

ML

MODEL

Precision measures how many positive predictions are right; recall measures how many actual positives are found.



OUT

EVIDENCE

Calculate both and choose a priority for two consequence scenarios.

QUESTION Calculate both and choose a priority for two consequence scenarios.

MODEL IDEA Precision measures how many positive predictions are right; recall measures how many actual positives are found.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Confusion counts: TP=8, FP=2, FN=4, TN=16. Precision = $8 \div (8+2) = 80\%$.

Draw a fresh data → model → prediction → evaluation trace:

A closer explanation

For matrix $\begin{bmatrix} 12 & 4 \\ 1 & 3 \end{bmatrix}$, TN=12, FP=4, FN=1, TP=3. There are seven predicted positives and four reference positives. Precision is $3/7$, about 42.9 percent. Recall is $3/4=75$ percent. Accuracy is $15/20=75$ percent. A correct denominator matters more than memorizing a percentage.

```
tn, fp, fn, tp = 12, 4, 1, 3

precision = tp / (tp + fp) if tp + fp else None
recall = tp / (tp + fn) if tp + fn else None
accuracy = (tn + tp) / (tn + fp + fn + tp)

print(precision, recall, accuracy)

# None means unavailable, not zero or perfect.
```

Trace the calculation on paper before running this optional Python check. Use the stated fictional inputs and compare intermediate values. The code checks this example; your independent workbook case uses different evidence.

Connect the picture and the explanation

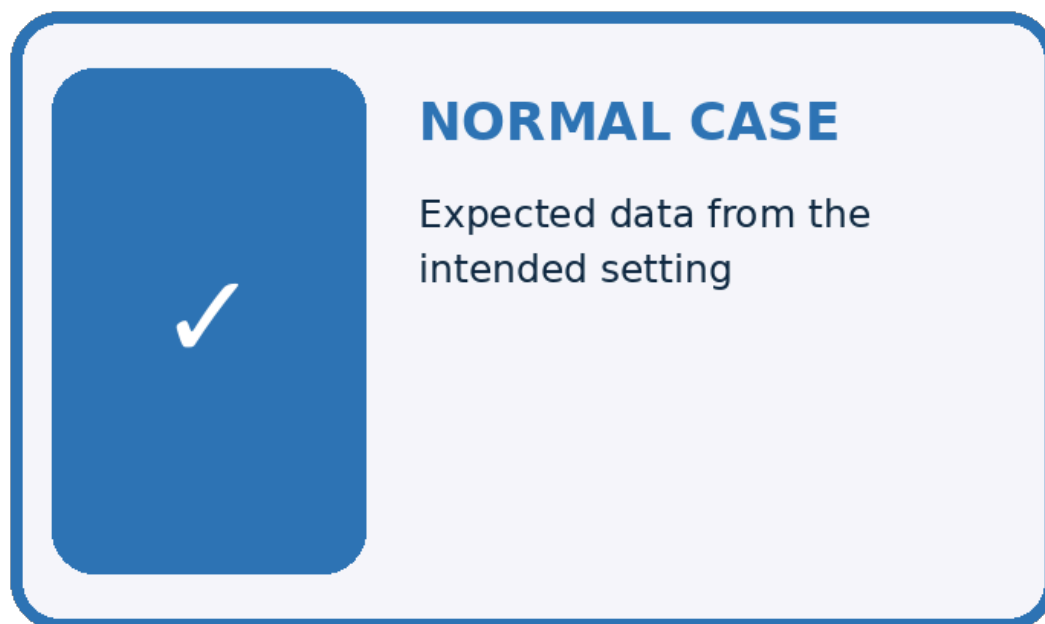
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

EVALUATION DESIGN

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.





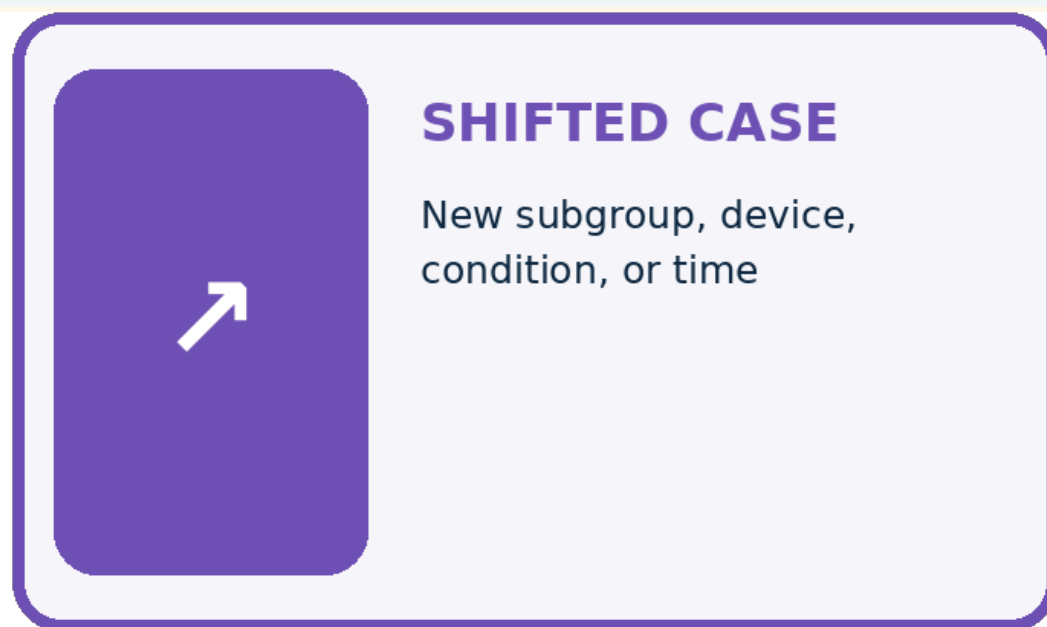
BOUNDARY CASE

Value near a threshold or class boundary



MISSING CASE

Unknown field, label, or necessary evidence



NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain prediction.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change: Correct two false negatives into true positives.

Prediction: Precision becomes $8/10=80\%$; recall becomes $8/10=80\%$.

Reason: TP becomes 8 and FN 2; FP remains 2, so the two denominators happen to equal 10.

5. Words that unlock the idea

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.

1

PRECISION

TP divided by TP plus FP

2

RECALL

TP divided by TP plus FN

3

FALSE ALARM

false-positive prediction

4

MISS

false-negative prediction

Term · Precise meaning · My ML example

Term: precision

Precise meaning: TP divided by TP plus FP

My ML example: _____

Term: recall

Precise meaning: TP divided by TP plus FN

My ML example: _____

Term: false alarm

Precise meaning: false-positive prediction

My ML example: _____

Term: miss

Precise meaning: false-negative prediction

My ML example: _____

MISCONCEPTION TO REPAIR Precision and recall are two names for the same accuracy calculation.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word · Meaning

Word: Precision

Meaning: True positives divided by all predicted positives

Word: Recall

Meaning: True positives divided by all reference positives

Word: Denominator

Meaning: The group counted in the bottom of a fraction

Word: Undefined

Meaning: A ratio with no valid value because its denominator is zero

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

WEEK 17 • WORKED EVIDENCE

- 1 Confusion counts: TP=8, FP=2, FN=4, TN=16.
- 2 Precision = $8 \div (8+2) = 80\%$.
- 3 Recall = $8 \div (8+4) \approx 67\%$.
- 4 Choose threshold and human review by comparing the cost of two false alarms with four misses.

STEP 1 Confusion counts: TP=8, FP=2, FN=4, TN=16.

STEP 2 Precision = $8 \div (8+2) = 80\%$.

STEP 3 Recall = $8 \div (8+4) \approx 67\%$.

STEP 4 Choose threshold and human review by comparing the cost of two false alarms with four misses.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

A closer explanation

Precision recall and denominators at a glance

Part · Evidence or action · Interpretation

Part: Precision

Evidence or action: 3 correct flags

Interpretation: 7 total flags

Part: Recall

Evidence or action: 3 found blocked cases

Interpretation: 4 reference blocked

Part: Accuracy

Evidence or action: 15 correct labels

Interpretation: 20 total records

Draw a box around the predicted-blocked column for precision and around the reference-blocked row for recall. Their overlap is TP. A model can flag nearly everything and find most blocked cases while producing many false alarms. Conversely, a model can flag very few cases accurately while missing many positives.

Calculate precision, recall and accuracy for the worked matrix. Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Precision and Recall

Put precision and recall denominators on the picture instead of memorising names.

The facts we will use

Positive means spotted. TP=6, FP=2, FN=4, TN=8 in twenty fictional leaves.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Keep precision and recall denominators distinct

	predicted spotted	predicted healthy
actual spotted	6	4
actual healthy	2	8

Read a cell using both its row and column labels.

Precision and Recall

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Predicted positives

Eight leaves were flagged: $\text{precision} = 6/(6+2) = 75\%$.



2. Actual positives

Ten leaves were spotted: $\text{recall} = 6/(6+4) = 60\%$.



3. Keep goals separate

Precision asks how reliable flags are; recall asks how many spotted leaves were caught.

CHANGE ONE THING

Correct two false negatives into true positives.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Predicted positives

What happens and why: Eight leaves were flagged: $\text{precision} = 6/(6+2) = 75\%$.

Step: 2. Actual positives

What happens and why: Ten leaves were spotted: $\text{recall} = 6/(6+4) = 60\%$.

Step: 3. Keep goals separate

What happens and why: Precision asks how reliable flags are; recall asks how many spotted leaves were caught.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Precision asks how reliable flags are; recall asks how many spotted leaves were caught. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

If a relevant denominator is zero, report that metric as undefined under the chosen convention rather than silently inventing success.

Read the labelled picture: Were the original precision and recall denominators the same? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Change one thing: Correct two false negatives into true positives. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



COMPARE • DIAGNOSE • REPAIR

Tempting claim: Precision and recall are two names for the same accuracy calculation. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: passes

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: defined behavior

Actual: _____

Likely cause / next test: _____

Evidence record: missing case

Expected: safe fallback

Actual: _____

Likely cause / next test: _____

Evidence record: shifted case

Expected: measured limit

Actual: _____

Likely cause / next test: _____

CLAIM TO CHALLENGE Precision and recall are two names for the same accuracy calculation.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Ratios describe this evaluation set, not a guarantee for an individual case. Library zero-division policies differ; document yours. Do not select a metric only after seeing which one makes a model look best.

A team reports recall 100 percent but hides precision and counts. What is missing?

8. Practise together

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



FREEZE THE EXPERIMENT

Record question, intended use, data version, grouped split, preprocessing, features, model settings, seed, metric, predictions, errors, subgroup results, and human review before claiming improvement.

Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/preprocessing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

With $[[16, 0], [4, 0]]$, there are no predicted positives. Precision is undefined because its denominator is zero; recall is $0/4=0$. With $[[16, 4], [0, 0]]$, recall is undefined because no reference positives were evaluated. Record counts and the undefined result instead of calling it perfect.

Practise with less help: Calculate both ratios

Use workbook W2 for the connected example “Precision recall and denominators”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Precision recall and denominators. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Were the original precision and recall denominators the same? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Metric Tradeoff Lab: compute both metrics for two thresholds, map errors to consequences, and defend a choice.

1. Write the hypothesis and expected evidence before running.
2. Metric Tradeoff Lab: compute both metrics for two thresholds, map errors to consequences, and defend a choice.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

Case: Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the example.



ML BUILDER MASTERY
1 Define • 2 Calculate • 3 Trace code • 4 Run experiment • 5 Diagnose errors • 6 Improve one cause • 7 Defend limits

DEFINE Use all four terms precisely: precision, recall, false alarm, miss.

CALCULATE Reproduce the worked evidence with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE Precision and recall are two names for the same accuracy calculation.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed evidence.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

11. Connect your project

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same evidence-based capstone.

**RESPONSIBLE PLANT CLASSIFIER**

Set a Plant Classifier review threshold and explain which error type it prioritizes and who makes the final decision.

THIS WEEK'S CHECKPOINT Set a Plant Classifier review threshold and explain which error type it prioritizes and who makes the final decision.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My evidence-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



INDEPENDENT ML BUILDER PROOF
Calculate both and choose a priority for two consequence scenarios.

REQUIRED PROOF Calculate both and choose a priority for two consequence scenarios.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent evidence and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

NEXT BRIDGE Carry the experiment evidence into: Error Analysis.

13. Study a complete case

Week 17 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Positive is spotted: TP=8, FP=2, FN=4.

The question

Compute precision and recall.

Worked explanation

Precision= $8/10=80\%$; recall= $8/12$ about 66.7%. Denominators are predicted positives and actual positives respectively.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Precision and recall use different denominators

Evidence or stage · Value or result

Evidence or stage: Precision, predicted positives

Value or result: 8/10

Evidence or stage: Recall, actual positives

Value or result: 8/12

Positive is spotted: TP=8, FP=2, FN=4.

Compute precision and recall.

Precision=8/10=80%; recall=8/12 about 66.7%. Denominators are predicted positives and actual positives respectively.

14. Try a new case independently

Independent case: Apply independently

Use [[8, 2],[3, 6]]. Give TP, FP, FN, precision, recall and accuracy.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 17 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

With disease positive, TP=6, FP=2 and FN=4. Calculate precision and recall, showing both denominators.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Set a Plant Classifier review threshold and explain which error type it prioritizes and who makes the final decision.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

TP=9,FP=3,FN=6,TN=12. Compute precision and recall. What happens to precision when a system predicts no positives?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: Put precision and recall denominators on the picture instead of memorising names.

By the end, I can · How I will show it

By the end, I can: Choose the denominator that answers the question

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Calculate precision and recall with a declared positive class

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Report undefined ratios honestly

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Choose the denominator that answers the question

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Put precision and recall denominators on the picture instead of memorising names.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Precision asks how reliable flags are; recall asks how many spotted leaves were caught.

- 1. Predicted positives
- 2. Actual positives
- 3. Keep goals separate

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Keep goals separate Precision asks how reliable flags are; recall asks how many spotted leaves were caught. Read the picture in order. Precision asks how reliable flags are; recall asks how many spotted leaves were caught. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking

the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: Correct two false negatives into true positives. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. Precision becomes $8/10=80\%$; recall becomes $8/10=80\%$.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Precision becomes $8/10=80\%$; recall becomes $8/10=80\%$. TP becomes 8 and FN 2; FP remains 2, so the two denominators happen to equal 10.

Repair a missing building block

Review Accuracy and Imbalance (ai-4/16). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why can the 90% baseline still be unsuitable for a detection task? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Scores and Calibration Intuition (ai-4/14). Goal: Interpret calibration using groups of forecasts. Short practice: Does a calibrated 0.8 score mean every such leaf is spotted? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Decision Trees (ai-4/11). Goal: Fit a root question and leaf predictions from training examples. Short practice: Which input distinguishes the original and changed tree? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first,

run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.